



Full Length Article

Signal processing and force control for precision machining with grinding system integrating an aerostatic suspension compact pneumatic actuator

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ABSTRACT

Signal processing is the core component of precision machining, where reliable contact force signals underpin monitoring, control, and quality assurance. Pneumatic actuators are highly favored in grinding applications due to their compliance, but friction and stick-slip phenomena distort force signals and degrade closed-loop performance. This study developed an aerostatic suspension compact pneumatic actuator (ASCPA) that suppresses friction at its source and compares it with traditional pneumatic actuators (TPA) and commercial low-friction pneumatic actuators (LFPA). Through piston-level fluid dynamics simulation, a balance between radial load capacity and air consumption was achieved, enabling prototype manufacturing and friction validation testing. From a manufacturing-oriented signal perspective, the adaptive state-space force signal recognition method EM-KF-RTS was proposed. This method integrates a Kalman filter with a Rauch-Tung-Striebel smoother and employs an expectation maximization algorithm for online parameter adaptation. Using only standard force values and air supply pressure measurements from the test bench, this estimator significantly enhances resistance to vibration, electromagnetic interference, and sensor noise, enabling precise force reconstruction. The actuator-algorithm stack was evaluated under constant, sinusoidal, and variable-amplitude frequency-modulated conditions, as well as during sudden external contact and inter-chamber pressure changes on the dynamic grinding test bench. Results demonstrate that the EM-KF-RTS method achieves stable force recognition under constant, sinusoidal, and variable amplitude-frequency reference signals, as well as during sudden external contact and inter-chamber pressure changes. It exhibits superior dynamic response performance, lower root mean square error, and stronger interference suppression capabilities.

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1. Introduction

Grinding and polishing processes are indispensable in precision manufacturing, particularly in fields such as optical lens fabrication, aeroengine blade finishing, and dental composite treatment, where stringent requirements on surface quality and dimensional accuracy must be satisfied [1–4]. Achieving uniform material removal and mitigating subsurface damage relies heavily on precise force–position control [2]. In this context, advanced control strategies—including impedance and neural network-based, fuzzy adaptive, and finite-time convergent methods—have been developed to improve trajectory tracking, enhance adaptability to complex geometries, and maintain process stability [5–10]. Pneumatic actuators have garnered considerable attention due to their intrinsic compliance, low cost, and capability for smooth force modulation, making them well-suited for safe human–robot interaction and versatile manipulation [11]. They have been widely applied in soft grippers, compliant robotic end-effectors, capsule polishing, blade grinding, and medical applications such as tissue stabilization, while also demonstrating scalability for dual-arm systems and hybrid fingers to achieve adaptive grasping [12–15]. Moreover, their ability to maintain constant normal force during polishing of unknown-model workpieces highlights their effectiveness in automated manufacturing scenarios where rigid alternatives often fail [16,17].

Despite these advantages, pneumatic systems face critical technical challenges that restrict their deployment in high-precision polishing. Chief among these are substantial friction, response delays, and inherent nonlinear dynamics, which collectively compromise positioning accuracy and force stability [18,19]. Friction, in particular, induces hysteresis and stick–slip effects, leading to significant errors in force estimation and trajectory tracking [19]. These limitations become even more pronounced in complex curved surface machining, such as turbine blades and diamond epitaxial layers, where fast and accurate real-time force regulation is required to avoid overstress and maintain contour fidelity [20,21]. Existing compensation methods, including adaptive observers, sliding-mode approaches, and PWM-based force control, have alleviated some nonlinearities and delays but remain insufficient for stringent polishing requirements [22–25]. Recent progress in robotic grinding also highlights that improving force control precision requires explicit treatment of nonlinear friction. In particular, Xun et al. [26] proposed a novel continuous dynamic friction model and a corresponding compensation method for force-controlled grinding actuators, which improves force regulation precision and grinding performance in experiments. This line of research complements friction-reduction-by-design efforts in pneumatic actuators by addressing friction-induced disturbances at the control and compensation level.

Currently, research has mainly focused on reducing friction in pneumatic actuators, improving control precision, and optimizing their static and dynamic performance. Whitney et al. [27] introduced a passive fluid transmission using antagonist pairs of rolling diaphragm cylinders, achieving near-zero stiction and complete back drivability due to a sealed fluid system. Qian et al. [28–30] employed high-frequency longitudinal vibration to reduce friction in pneumatic cylinders, achieving up to a 46.6% reduction in dynamic friction and improving trajectory tracking accuracy [29]. Liu et al. [31] proposed a model-free adaptive control strategy for a frictionless cylinder-driven pneumatic gravity compensation system, achieving a steady-state error below 200 Pa. Zhu et al. [32] developed a frictionless pneumatic cylinder with an aerostatic bearing piston, optimized by finite element analysis for radial load-carrying capacity and stability. Related studies on aerostatic and hydrostatic bearings have revealed that recess geometry, restrictor design, and supply pressure significantly influence performance, load capacity, and stability [33–39].

However, unwanted vibrations often occur during grinding and polishing, leading to reduced efficiency and potential structural damage. Tang et al. [40] investigated vibration suppression of a series elastic actuator for robotic grinding by integrating a reconstructed hybrid optimized input shaper into a multi-feedforward frame and reported improvements in force tracking and grinding quality. Characterizing the contact force in the tool–workpiece interaction zone is critical, as accurate identification of this force not only suppresses vibrations but also prevents machining damage. Researchers have explored various inverse techniques for force identification in structural dynamics, often leveraging advanced signal processing, time–frequency analysis, and non-contact computer vision methods for comprehensive vibration measurement and analysis [41–43]. For example, Khoo et al. [44] applied operating deflection shape analysis combined with frequency response function measurements and pseudo-inverse methods to recover impact forces on lightweight structures in under-determined, even-determined, and over-determined cases. Lourens et al. [45] proposed an augmented Kalman filter that incorporates modeling errors to estimate unknown forces and states from acceleration data. Azam et al. [46] introduced a dual Kalman filter for joint input–state estimation using output-only accelerations, addressing drift in force and displacement estimates. Naets et al. [47] improved Kalman-based estimation stability by adding dummy position measurements, overcoming limitations of acceleration-only data. Zhang et al. [48] developed a Bayesian framework for force reconstruction under model uncertainty and measurement noise using Monte Carlo Markov chains. Nayek et al. [49] formulated a Gaussian process latent force model in an augmented state-space representation for stochastic joint input–state estimation, demonstrating robustness against drift. Nguyen et al. [50] presented a pose-dependent optimal control method based on Gaussian process regression for active vibration suppression in robotic milling under periodic forces. Guo et al. [51] introduced active contact robotic milling to improve stability by controlling additional interface forces, suppressing chatter while maintaining flexibility and removal rates. Khan et al. [52] proposed a singular perturbation-based adaptive integral sliding mode control with state observers for flexible joint robots, enabling robust tracking under uncertainties and disturbances.

Although prior studies have improved pneumatic actuators and advanced inverse force identification, significant challenges remain—most notably simplified modelling, sensitivity to measurement noise, and limited robustness under dynamic contact conditions. To address these issues, ASCPA is developed together with an adaptive state-space estimation algorithm that integrates a Kalman filter, Rauch–Tung–Striebel smoother and expectation–maximization (EM–KF–RTS). A dynamic surface-grinding testbed is constructed to emulate practical manufacturing scenarios. On this platform, the ASCPA is compared with a TPA and a LFPA, and the force-identification capability of EM–KF–RTS is evaluated under representative references and disturbances. The main contributions are summarized as follows:

- 1) An ASCPA based on static-pressure air lubrication is designed. Piston parameters governing radial load capacity and air consumption are optimized via fluid-dynamic simulations to guide prototype fabrication. A dedicated friction rig quantifies tribological behavior against a TPA and an LFPA: relative to the LFPA, the ASCPA reduces maximum static friction and Coulomb friction by 99.12% and 98.56%, respectively; reductions reach 99.58% and 99.48% relative to the TPA.
- 2) An EM-KF-RTS framework is introduced for manufacturing-oriented force estimation under pervasive disturbances. By combining Kalman filtering, RTS smoothing and EM-based parameter adaptation, the method provides stronger noise suppression and more accurate force reconstruction than conventional frequency-domain approaches; simulations further demonstrate robustness for extracting dynamic forces at low signal-to-noise ratios.
- 3) A dynamic surface-grinding platform is used to assess both the actuator and the estimator under three undisturbed references (constant, sinusoidal and variable amplitude–frequency forces) and two disturbance scenarios (abrupt external contact and inter-chamber/supply-pressure variation). Results show that EM-KF-RTS consistently achieves reliable force identification across all conditions, while the ASCPA delivers faster response and higher tracking accuracy than the TPA and LFPA, thereby improving the analyzability of contact-force signals for precision grinding.

2. Actuation setup and friction characterization

This section provides a detailed exposition of the ASCPA. Initially, the design scheme of the ASCPA is presented. Subsequently, fluid dynamics simulations are conducted on the critical component, the aerostatic piston, to optimize its structural parameters, leading to the fabrication of a prototype. Finally, a friction test platform is established to evaluate the frictional characteristics of the ASCPA, with comparisons made against two pneumatic actuators, confirming its superior low-friction performance.

2.1. Actuator overview

Pneumatic end effectors used in grinding processes need to simultaneously possess characteristics such as low friction, rapid dynamic response, and high-precision force control, while maintaining a compact and practical mechanical structure. These requirements are particularly important in grinding force regulation, as friction-induced hysteresis, stick–slip behavior, and transmission nonlinearity directly reduce force tracking accuracy and distort the measured force signal. To address these issues, this paper develops an air-floating compact pneumatic actuator (ASCPA) based on the principle of hydrostatic air lubrication. In traditional pneumatic actuators, friction mainly originates from two contact interfaces: (i) the sliding contact between the piston and the cylinder wall; and (ii) the interaction between the piston rod and the front cover seal and guide components. The frictional effects caused by these contacts lead to nonlinear force transmission, reduced response sensitivity, and decreased stability in precision grinding applications. To suppress this friction at its source, the proposed ASCPA introduces an air-floating piston with surface orifices that generate a high-pressure air film, thereby achieving non-contact or near-non-contact levitation motion within the cylinder. Furthermore, an air-floating bearing is integrated into the front cover to further reduce friction between the piston rod and the front support structure.

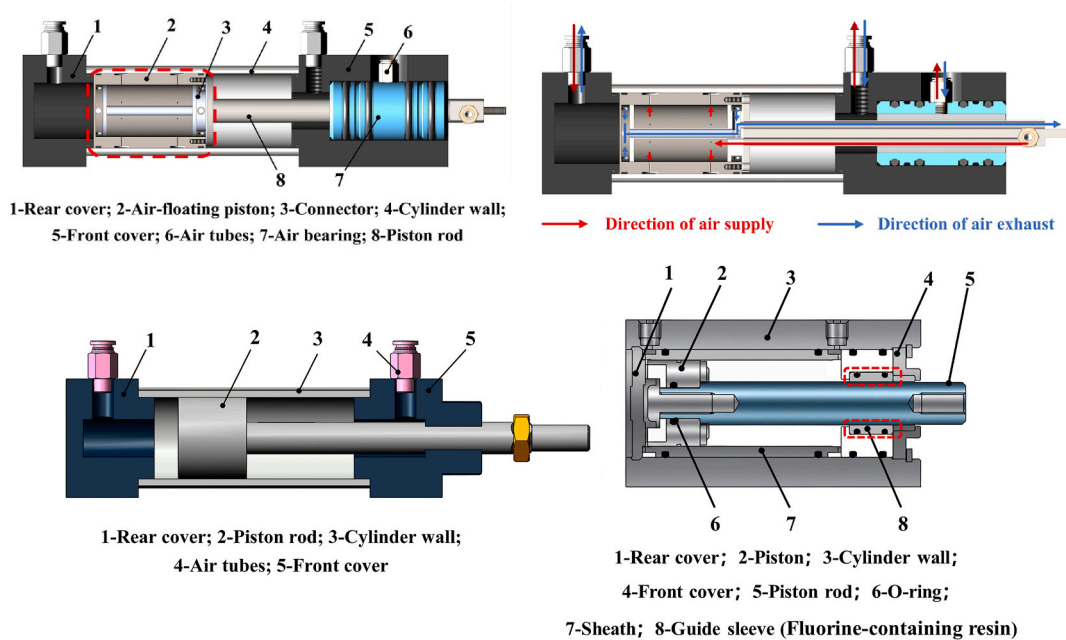


Fig. 1. Structural schematics of the three pneumatic actuators used in this study: (a) proposed ASCPA, (b) working principle of ASCPA, (c) traditional pneumatic actuator (TPA), and (d) low-friction pneumatic actuator (LFPA, SMC MQQT30-30D).

Compared to traditional contact-guided designs, this structure shifts the friction reduction strategy from “reducing friction at the solid contact interface” to “replacing solid contact with air-bearing support.” Fig. 1(a) shows the structural configuration of the proposed ASCPA, and Fig. 1(b) illustrates its working principle. As shown in Fig. 1(a), the ASCPA consists of a rear cover, an air-bearing piston, a connector, a cylinder wall, a front cover, air pipes, an air-bearing bearing (NEWWAY, S300601), and a piston rod. The air-bearing piston is hollow inside, allowing for independent air supply and exhaust, thereby establishing and stabilizing a high-pressure air film to achieve smooth suspension motion. The piston is connected to the piston rod via a central connector and secured by circumferentially distributed bolts. As shown in Fig. 1(b), compressed air flows through two independent channels within the piston rod and connector, respectively for supplying and exhausting the air-bearing piston; the red and blue arrows indicate the directions of air supply and exhaust, respectively. This structure maintains a uniform and stable air film between the air-bearing piston and the cylinder wall, while isolating the pressure within the two working chambers of the ASCPA. An independently supplied air bearing within the front cover provides low-friction support and guidance for the piston rod. Major structural components are constructed of 304 stainless steel to ensure mechanical strength and corrosion resistance.

To provide a clearer comparison with the actuators used in the experiment, Fig. 1(c) and (d) show schematic diagrams of a conventional pneumatic actuator (TPA) and a low-friction pneumatic actuator (LFPA), respectively. The TPA in Fig. 1(c) employs a conventional piston-cylinder structure without a dedicated friction-reducing mechanism. Therefore, its force transmission relies on direct solid-state contact sliding at the piston-cylinder and rod-guide interfaces, which typically introduces significant static friction, Coulomb friction, and friction-induced hysteresis. The LFPA in Fig. 1(d) is a commercially available low-friction cylinder (SMC, MQQT30-30D) used in the comparative test. As shown in the marked guide area, its friction reduction is primarily achieved through material lubrication design, using a fluoropolymer guide sleeve to reduce friction at the contact interface. In other words, the LFPA still maintains contact-guided motion; its friction reduction is mainly achieved by reducing the interfacial friction coefficient and improving lubrication conditions, rather than introducing a film support. As can be seen from the structural diagram, the proposed ASCPA differs fundamentally from TPA and LFPA in its friction reduction mechanism. TPA does not include a dedicated friction-reducing structure, while LFPA primarily reduces friction by using low-friction materials and lubrication at the solid-contact guide interface. In contrast, ASCPA utilizes film support to alter the contact state itself. With a supply of compressed air, a hydrostatic film forms in the bearing clearance around the air-bearing piston, generating a supporting pressure distribution that provides radial load-bearing capacity and maintains the separation between solid surfaces within the designed clearance range. This film support and separation effect is the theoretical basis for ASCPA's friction reduction. Because direct solid–solid contact is significantly reduced or avoided, stick–slip is eliminated, thus achieving a friction-reducing effect. Compared to TPA and LFPA, the proposed ASCPA provides a more favorable structural basis for precision grinding force control. Compared to TPA, it introduces specialized friction suppression designs at both the piston-cylinder interface and the rod-guide interface. Compared to LFPA, its main advantage lies in achieving friction reduction through air-bearing support and contact state transitions, rather than solely relying on low-friction contact materials and lubrication. This design is expected to reduce friction-induced hysteresis, improve motion sensitivity, and enhance force transmission consistency, thereby facilitating high-precision force control and reliable force signal recognition in grinding applications.

2.2. Research on the fluid characteristics of air-floating piston

The air-floating piston is central to the performance and stability of the ASCPA, particularly when coupled with grinding tools under varying loads. Radial disturbances induce eccentricity between the piston surface and cylinder wall, producing nonuniform air film thickness and pressure distribution, this generates a radial load-carrying force [53,54]. To ensure adequate load capacity with reasonable air consumption, the piston diameter was fixed at 20 mm with an aspect ratio of $L/D = 1.5$, following prior optimization studies [55]. An annular pressure-relief groove, designed on labyrinth sealing and pressure equalization principles, was incorporated to improve end-face pressure distribution and radial load capacity [56]; its width and depth were set to 1.5 mm and 0.5 mm, respectively, considering manufacturability and structural integrity.

The piston's radial load capacity and air consumption depend on parameters including average air film thickness, eccentricity, orifice diameter, supply pressure, and axial spacing of orifices. Numerical simulations using ANSYS Fluent were conducted to evaluate

Table 1
Default simulation parameters of air floating piston.

Parameters	Symbols	Values
Outer diameter of air-floated piston	D_p/mm	20
Length of air-floated piston	L_p/mm	30
Diameter of pressure equalization chamber	D/mm	4
Depth of pressure equalizing chamber	H/mm	0.5
Width of annular pressure relief groove	W/mm	1.5
Depth of annular pressure relief groove	d/mm	0.5
Number of circumferential orifices	n	10
Number of axial orifice rows	m	2
Supply pressure	P/MPa	0.4
Average air film thickness	$h_0/\mu m$	30
Orifice diameter	d_0/mm	0.3
Eccentricity	e	0.4
Axial distance between Orifice	l/mm	15

their influence and identify critical structural parameters. Exploiting geometric symmetry, a three-dimensional half-domain flow model was established, with default values summarized in Table 1. Unless otherwise specified, these parameters were adopted in all simulations. Pressure distributions of the air film were extracted through CFD-Post, converted into the appropriate coordinate system in MATLAB, and visualized in three dimensions, as shown in Fig. 2. For small-hole throttling hydrostatic lubrication, the dominant parameters are the orifice diameter and average air film thickness. The latter governs the local pressure gradient, while the former controls mass flow rate and inlet momentum, jointly shaping pressure peaks and leakage distribution. As shown in Fig. 3, increasing the orifice diameter from 0.1 to 0.7 mm produces a non-monotonic variation in radial load capacity, with a clear optimum. With thicker films, this optimum shifts toward larger diameters, since reduced flow resistance requires a wider orifice to maintain favorable pressure distribution. Considering performance and manufacturability, the orifice diameter was set to 0.3 mm.

Fig. 4 shows that, at a given average air film thickness (h_0), increasing eccentricity (e) enhances load capacity by inducing asymmetric pressure distributions and higher local peaks. For a fixed eccentricity, load capacity first rises and then declines with increasing film thickness: moderate thickness promotes pressure buildup, whereas excessive thickness weakens the pressure gradient. Air consumption increases with both parameters, though the growth rate tapers due to pressure drop limits. Based on this trade-off, the average air film thickness was selected as 30 μm .

The axial distance between orifices (l) defines the effective axial region of air supply, while the number of circumferential orifices (n) determines the discreteness of the supply and facilitates pressure equalization. Together, they govern the pressure uniformity in the load-bearing zone. As shown in Fig. 5, for a fixed number of circumferential orifices, the radial load capacity first increases and then decreases with increasing axial distance: a moderate distance promotes pressure development, whereas excessive spacing hinders pressure recovery and reduces capacity. A short axial distance, combined with a greater number of circumferential orifices, generates a more uniform and higher-pressure field, thereby enhancing both the magnitude and uniformity of the radial load capacity. Accordingly, the axial distance was set to 15 mm and the number of circumferential orifices to 10.

2.3. Friction test of ASCPA

The ASCPA was machined according to the structural parameters defined above. Ideally, friction in the ASCPA arises primarily from air viscous drag; however, manufacturing and assembly imperfections can introduce additional friction beyond theoretical predictions. To ensure that the machined ASCPA achieves high-precision control, we evaluated friction using the LuGre friction model with experimental parameter identification.

The LuGre model is expressed as [57]:

$$f = \sigma_0 z + \sigma_1 \frac{dz}{dt} + \sigma_2 v \quad (1)$$

where σ_0 , σ_1 , and σ_2 are friction coefficients representing bristle stiffness, bristle damping, and viscous friction, respectively.

The differential equation for the average bristle deflection z is:

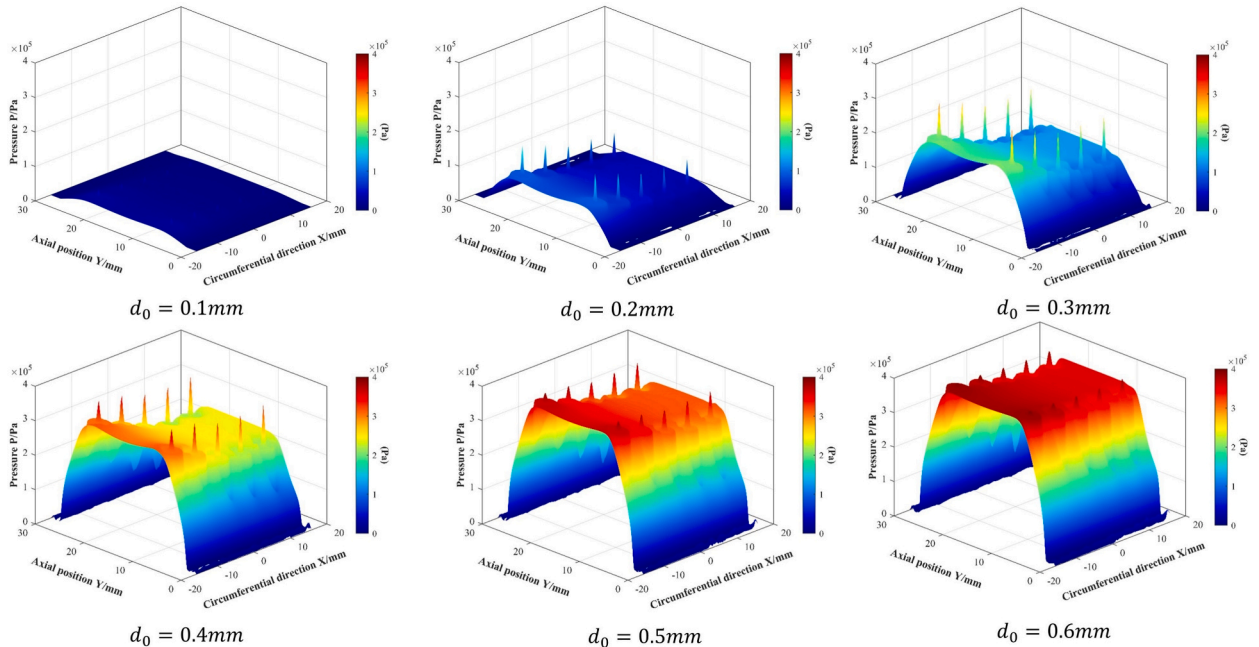
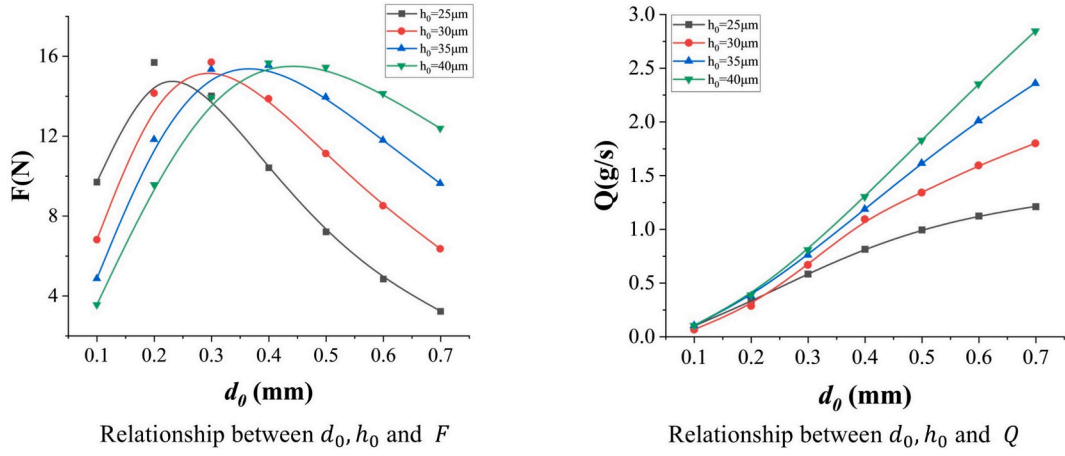
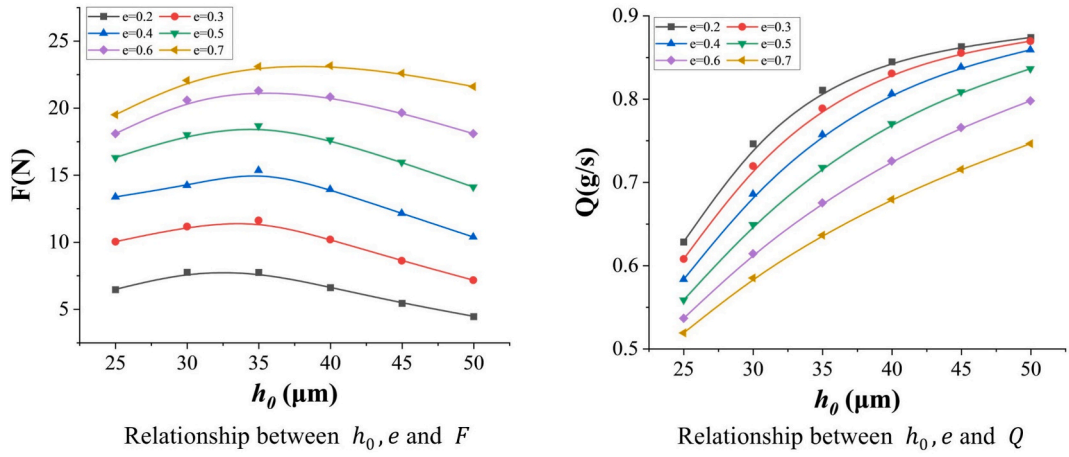
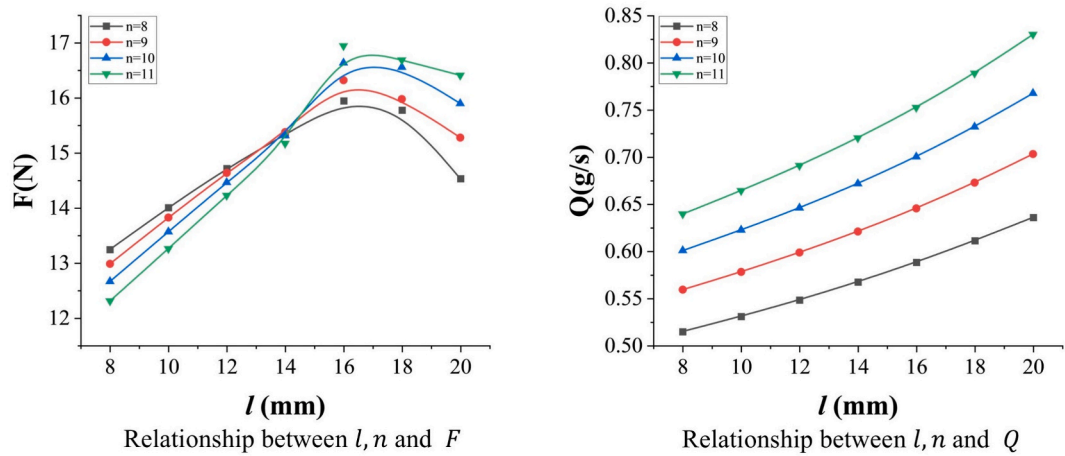


Fig. 2. 3D visualization of the pressure distribution in the air film with different orifice diameters.

Fig. 3. The simulation results for different d_0 and h_0 .Fig. 4. The simulation results for different h_0 and e .Fig. 5. The simulation results for different l and n .

$$\frac{dz}{dt} = v - \frac{\sigma_0 z |v|}{F_c + (F_s - F_c) e^{-(v/v_s)^2}} \quad (2)$$

where F_c is the Coulomb friction, F_s is the maximum static friction, v_s is the Stribeck velocity.

Under steady motion ($dz/dt = 0$), the model reduces to:

$$f = \left(F_c + (F_s - F_c) e^{-(v/v_s)^2} \right) \text{sign}(v) + \sigma_2 v \quad (3)$$

Because $\text{sign}(v)$ is non-differentiable, we replace $\text{sign}(v)$ with $2\arctan(15v)/\pi$ to improve agreement with measured data. To obtain parameter values, we fitted the friction curve experimentally and performed parameter identification.

To compare performance, the ASCPA was tested alongside a TPA (Airtac) and a commercially available LTPA (SMC MQQT20). As shown in Fig. 6, the piston rod was fixed, and a stepper motor drove a ball-screw slide at constant speed to ensure uniform extension and retraction. A force sensor (Zhongnuo Transducer, ZNLBS-V1) measured the cylinder friction while the control cabinet operated the ball-screw slide at various constant velocities. The procedure was repeated for each pneumatic actuator. The resulting friction-velocity curves were fitted, and the parameters were identified using Eq. (3). The fitted curves and identified parameters are presented in Fig. 7 and Table 2.

Due to the use of conventional contact seals, the piston in the TPA exhibits high stick-slip and large Coulomb/static friction. The LFPA reduces these effects through lower-friction sealing and improved guidance, but still relies on contact sealing and thus retains significant friction. In contrast, the ASCPA employs hydrostatic air suspension and a non-contact air bearing for rod guidance, forming a stable air film that eliminates solid-solid contact and markedly lowers both coulomb and static friction. Compared to LFPA, the ASCPA reduces maximum static friction by 99.12% and Coulomb friction by 98.56%. Compared to TPA, the reductions are 99.58% and 99.48% for maximum static and Coulomb friction, respectively.

3. Dynamic grinding platform and data acquisition

This section presents an adaptive state-space estimation algorithm (EM-KF-RTS), which integrates a Kalman filter with a Rauch-Tung-Striebel smoother and employs EM-based iterative optimization to adaptively identify model parameters for optimal signal estimation. To verify its effectiveness, a dynamic surface grinding force-control platform was constructed, integrating ASCPA, LFPA, and TPA to emulate practical grinding scenarios.

Fig. 8 illustrates the overall measurement-estimation-control flow implemented on the dynamic grinding force control platform. During the grinding process, the pneumatic actuator generates a contact force at the interface between the end effector and the workpiece, and the force sensor outputs the corresponding raw measurement value $F_m(k)$. This measurement signal is directly input to the proposed EM-KF-RTS force identification module, which performs state-space optimal estimation to suppress noise contamination and periodic disturbances in the measurement, and generates a denoised force estimate $\hat{F}(k)$. The estimated force $\hat{F}(k)$ is then used as the feedback signal for the force regulation loop. The desired force command $F_d(k)$ serves as the reference input, forming a tracking error $e_F(k) = F_d(k) - \hat{F}(k)$ at the summation node. The force controller uses $e_F(k)$ as the regulation variable to calculate the control input $u(k)$ as the valve input signal command to drive the pneumatic actuator, thereby regulating the grinding force and closing the feedback loop. In this implementation, the control law follows reference [58], and the EM-KF-RTS module provides a force feedback signal for noise suppression to the controller, replacing the direct feedback based on $F_m(k)$ in the traditional implementation.

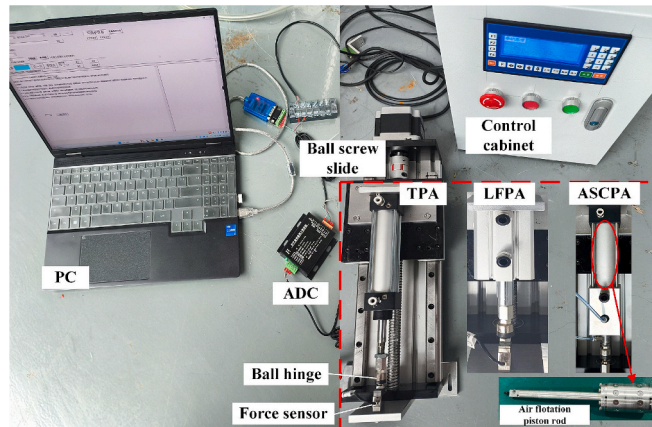


Fig. 6. Pneumatic actuator friction test platform.

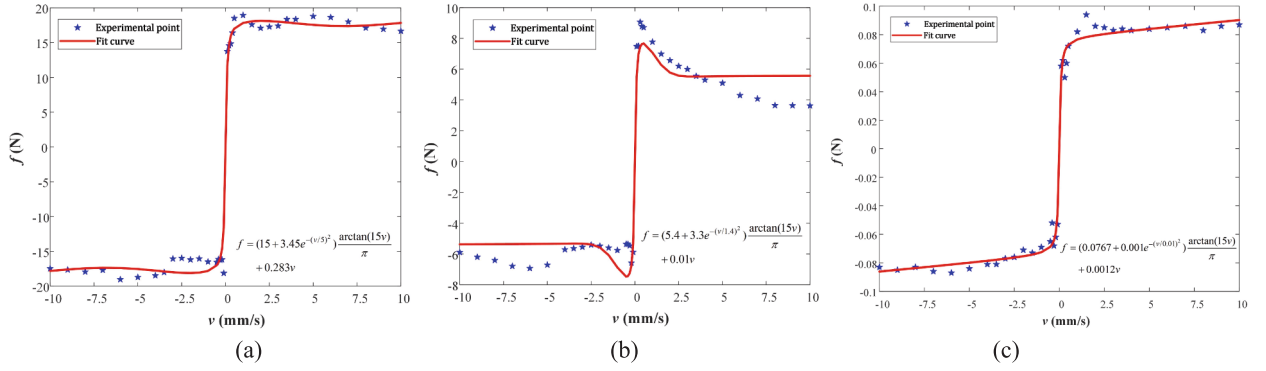


Fig. 7. Friction force fitting curve: (a) TPA; (b) LFPA; (c) ASCPA.

Table 2

Identification of friction parameters.

Parameters	Identification value		
	TPA	LFPA	ASCPA
F_c (N)	15	5.4	0.0777
F_s (N)	18.45	8.7	0.0767
v_s (mm/s)	5	1.4	0.01
σ_2 (N · s/mm)	0.283	0.01	0.0012

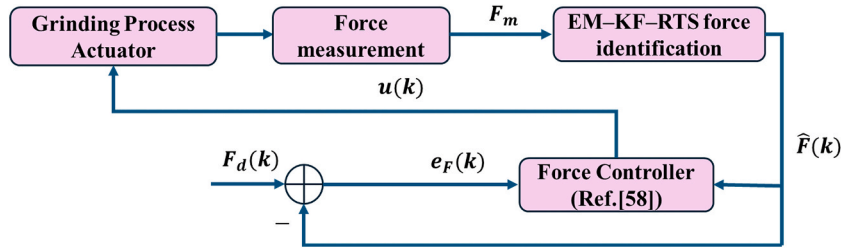


Fig. 8. Overall schematic of the measurement–estimation–control pipeline for grinding force regulation.

3.1. Grinding force signal identification

In precision grinding, the measured contact-force signal is non-stationary and noise-contaminated. Reliable force-signal identification is therefore a prerequisite for manufacturing-oriented signal processing, enabling meaningful time-domain indices and frequency-specific attenuation metrics that reflect process quality and robustness. However, due to machine vibration, electromagnetic interference, and intrinsic sensor noise, the measured force signal $F_m(t)$ is often heavily contaminated by noise. It can be modeled as a linear superposition of the true grinding force $F_{\text{true}}(t)$, multiple periodic components $F_{p,i}(t)$ caused by roller rotation and other factors, and broadband random noise $v(t)$:

$$F_m(t) = F_{\text{true}}(t) + \sum_{i=1}^{N_f} F_{p,i}(t) + v(t) \quad (4)$$

where N_f denotes the number of dominant periodic components.

To accurately extract dynamic force information closely related to the grinding state under strong noise, an adaptive state-space optimal estimation algorithm (EM–KF–RTS) is proposed, which integrates the Kalman filter (KF) with the Rauch–Tung–Striebel (RTS) smoother. In contrast to conventional methods relying on prior assumptions or direct frequency analysis, the proposed framework employs a data-driven iterative optimization strategy based on the Expectation–Maximization (EM) algorithm to adaptively identify key state-space parameters, thereby enabling optimal signal estimation.

3.1.1. Dynamic state-space model

Since the dominant periodic force components in grinding can be approximated as a superposition of harmonic oscillations, we construct a high-dimensional state-space model to describe their dynamics. For a harmonic component with angular frequency ω_i , its

dynamics can be fully represented by a pair of orthogonal states (sine and cosine terms). Accordingly, the discrete-time state vector $\mathbf{x}_k \in \mathbb{R}^{(2N_f+1) \times 1}$ at time k for a system with N_f harmonic components and a DC offset is defined as:

$$\mathbf{x}_k = [s_{1,k}, c_{1,k}, s_{2,k}, c_{2,k}, \dots, s_{N_f,k}, c_{N_f,k}, d_k]^T \quad (5)$$

where $s_{i,k}$ and $c_{i,k}$ represent the sine and cosine states of the i -th frequency component, d_k denotes the DC offset.

The corresponding discrete-time state-space model is given by:

$$\begin{cases} \mathbf{x}_k = \mathbf{A} \mathbf{x}_{k-1} + \mathbf{w}_k \\ y_k = \mathbf{H}_k \mathbf{x}_k + v_k \end{cases} \quad (6)$$

where y_k denotes the scalar measurement at time k , $\mathbf{w}_k \sim \mathcal{N}(0, \mathbf{Q})$ and $v_k \sim \mathcal{N}(0, R)$ represent the process noise and measurement noise, respectively.

The state transition matrix $\mathbf{A} \in \mathbb{R}^{(2N_f+1) \times (2N_f+1)}$ is block-diagonal:

$$\mathbf{A} = \text{diag}(\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_{N_f}, 1) \quad (7)$$

where each 2×2 block $\mathbf{A}_i$ is a rotation matrix describing the one-step evolution of the i th harmonic component:

$$\mathbf{A}_i = \begin{bmatrix} \cos(\omega_i \Delta t) & \sin(\omega_i \Delta t) \\ -\sin(\omega_i \Delta t) & \cos(\omega_i \Delta t) \end{bmatrix} \quad (8)$$

The observation matrix $\mathbf{H}_k \in \mathbb{R}^{1 \times (2N_f+1)}$ maps the unobservable state vector $\mathbf{x}_k$ to the measurable output y_k . It is formed from the amplitudes $\{Amp_i\}$ of the frequency components and the DC offset coefficient:

$$\mathbf{H}_k = [0, Amp_1, 0, Amp_2, \dots, 0, Amp_{N_f}, 1] \quad (9)$$

where Amp_i denotes the amplitude of the i -th component.

3.1.2. EM-based adaptive parameter estimation

The unknown parameters $\theta = \mathbf{H}, \mathbf{Q}, R$ are identified iteratively using the EM algorithm, which alternates between state estimation and parameter update until convergence. The procedure is shown in pseudo-code in [Algorithm 1](#).

Algorithm 1. (Pseudo-code of the EM-KF-RTS algorithm)

Input: Measured signal $y_k, k = 1^N$, sampling interval Δt
Output: Optimal model parameters θ_{opt} , estimated signal $\hat{F}(t)$
1: Perform FFT on y_k to estimate dominant frequencies f_i and construct state-transition matrix $\mathbf{A}$;
2: Initialize parameters $\theta^{(0)} = \mathbf{H}^{(0)}, \mathbf{Q}^{(0)}, R^{(0)}$;
3: Set iteration index $j = 0$;
4: while (convergence not reached) do
5: E-step: Run KF-RTS with $\theta^{(j)}$ to obtain $\hat{\mathbf{x}}_{k N}, \mathbf{P}_{k N}, \mathbf{P}_{k,k-1 N}$;
6: M-step: Update $\theta^{(j+1)} = \mathbf{H}, \mathbf{Q}, R$ using closed-form formulas;
7: $j = j + 1$;
8: end while
9: Run final KF-RTS with θ_{opt} to reconstruct $\hat{F}(t)$.

3.1.3. Optimal state estimation and signal reconstruction

After obtaining the optimal model parameters $\theta_{\text{opt}} = \mathbf{H}_{\text{opt}}, \mathbf{Q}_{\text{opt}}, R_{\text{opt}}$ through EM. The smoothed grinding force signal $\hat{F}(t)$ is reconstructed as:

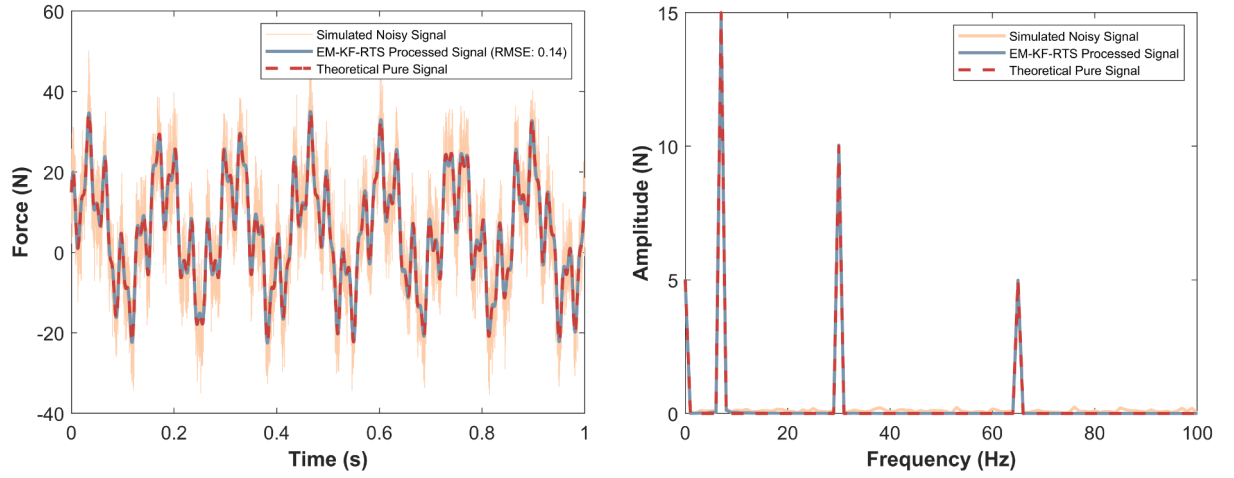
$$\hat{F}(k\Delta t) = \mathbf{H}_k \hat{\mathbf{x}}_{k|N} \quad (10)$$

This framework effectively suppresses broadband random noise while maintaining strong consistency with the dynamic characteristics of the signal, thereby preserving the essential features of grinding mechanics.

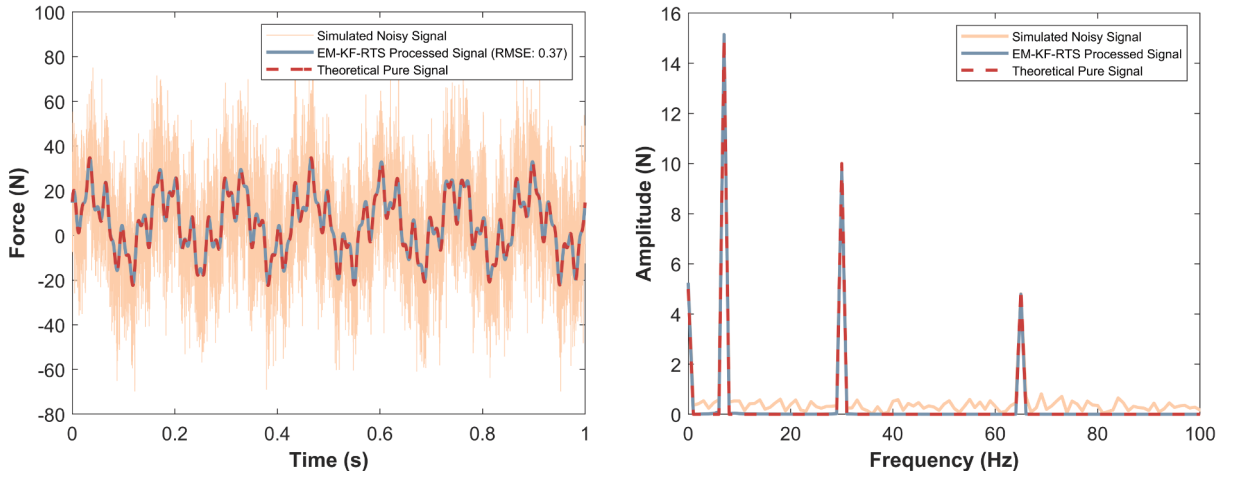
3.2. Simulation of grinding force identification under different noise levels

In practical engineering measurements, noise contamination is unavoidable and can adversely affect the accuracy of any identi-

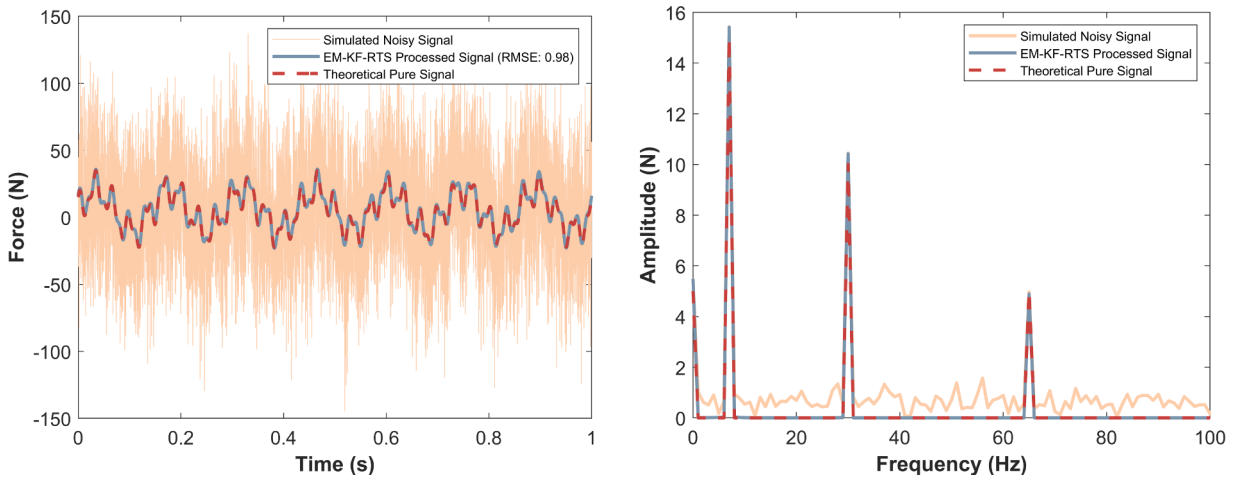
fication method. To assess the noise resistance of the proposed EM-KF-RTS approach, a multi-sinusoidal grinding force defined in Eq. (11) was used as the reference signal, and different noise levels were added to emulate real operating conditions. The total simulation time was set to 1 s with a sampling rate of 10000 Hz.



(a) Recognition results of force signals with 10% noise



(b) Recognition results of force signals with 30% noise



(c) Recognition results of force signals with 60% noise

Fig. 9. Comparison of force signal identification under different noise levels.

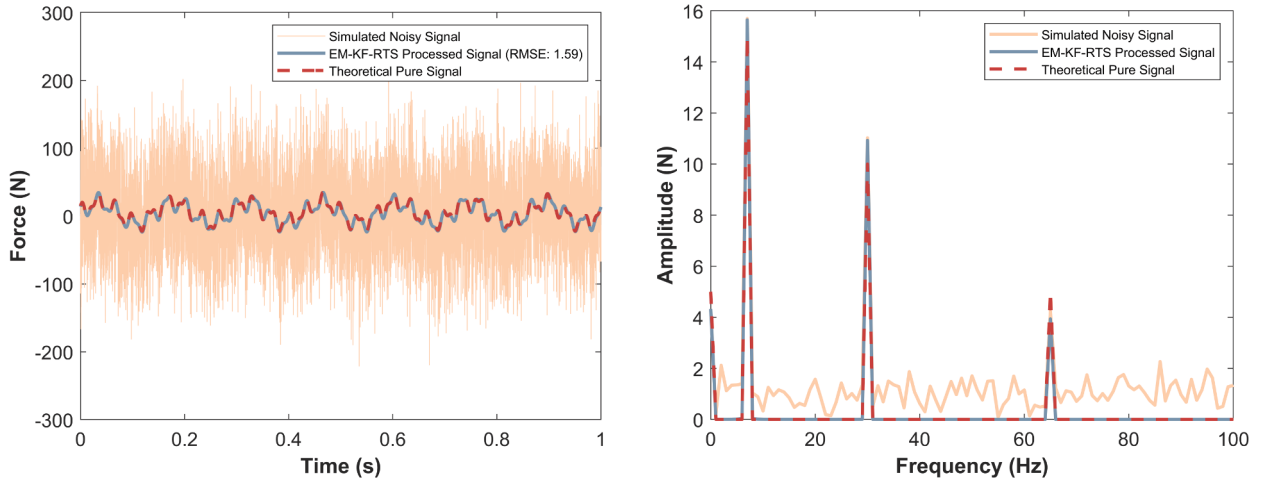


Fig. 9. (continued).

$$F(t) = 5 + 15\sin(14\pi t) + 10 \cdot \cos(60\pi t) + 5 \cdot \sin(130\pi t) \quad (11)$$

As shown in Fig. 9, force signals with 10%, 30%, 60%, and 100% noise levels were selected. The simulation results demonstrate that, when comparing the root mean square error (RMSE) performance of EM-KF-RTS under different noise conditions, the RMSE variation increases with higher noise content. This is attributed to the stronger disturbance introduced by higher noise levels. Nevertheless, the

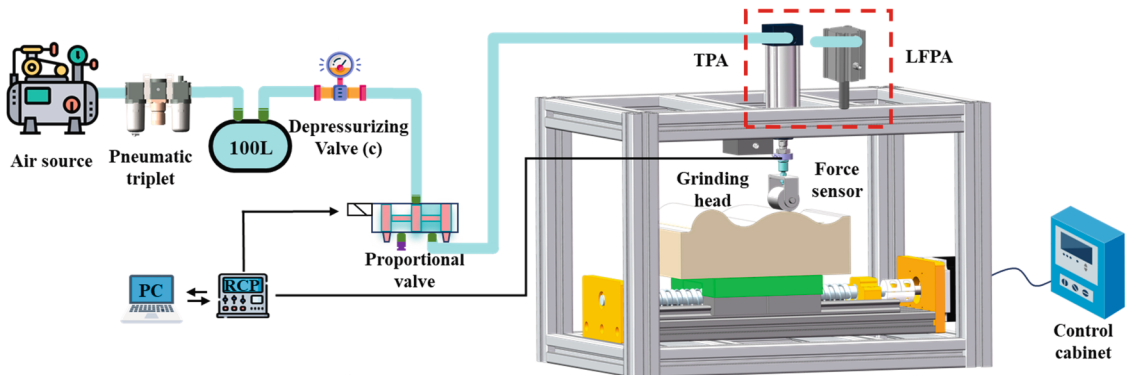
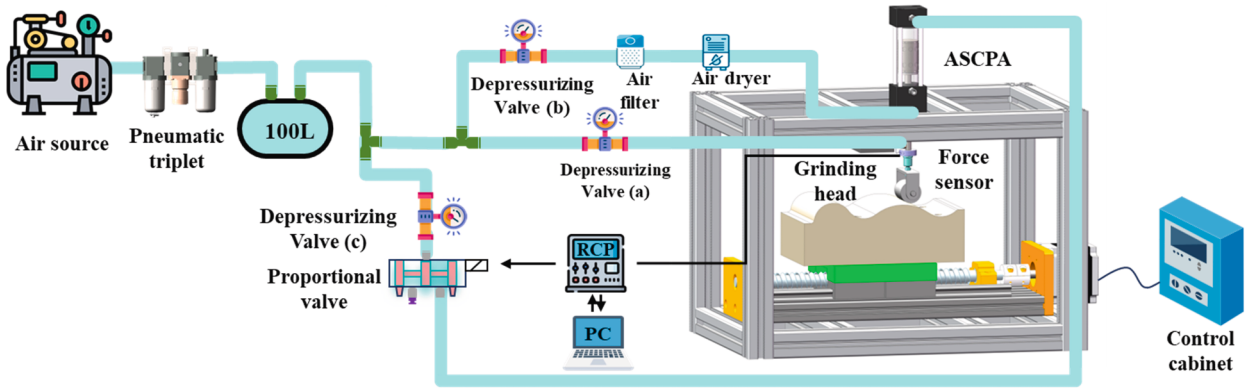


Fig. 10. The schematic diagram of dynamic surface grinding.

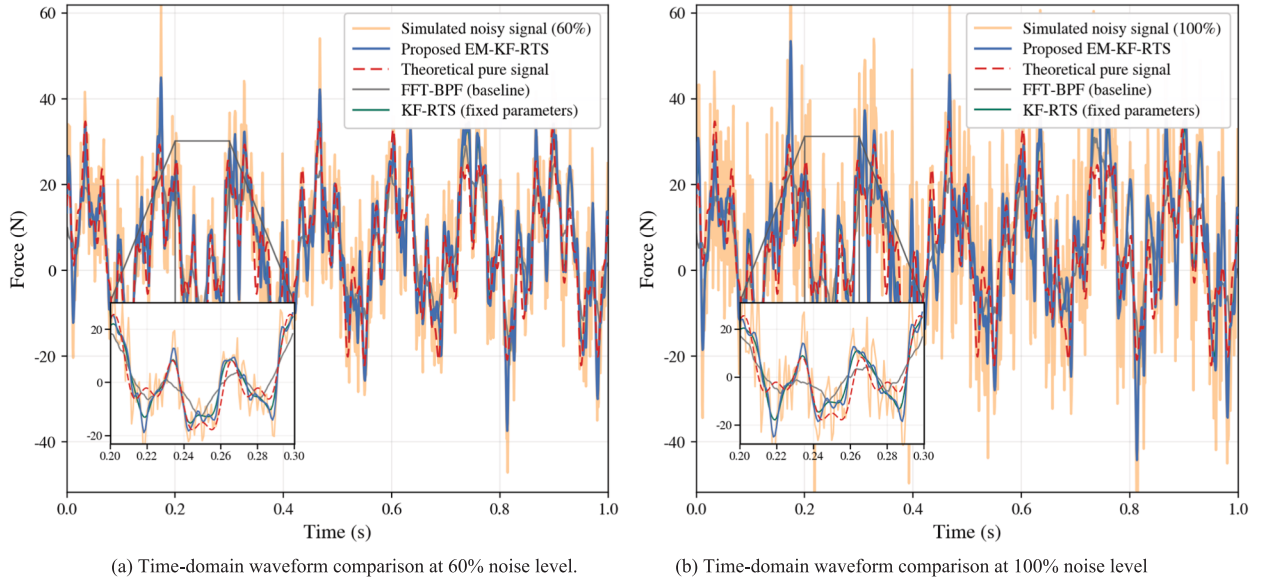


Fig. 12. Dynamic surface grinding test experimental platform.

Table 3

RMSE comparison of force-identification methods at representative noise levels.

Method	RMSE at 60% noise (N)	RMSE at 100% noise (N)
FFT-BPF (baseline)	2.18	3.52
KF-RTS (fixed parameters)	1.46	2.41
EM-KF-RTS (Proposed)	0.98	1.63

increases with increasing noise level, the proposed method maintains the best reconstruction accuracy and the least performance degradation trend among all the comparison methods, demonstrating stronger robustness under severe noise pollution.

3.4. Signal processing protocol & metrics

For a constant force signal, identification performance is assessed using the steady-state standard deviation (σ) and the peak-to-peak ripple (F_{pp}). The steady-state standard deviation reflects random fluctuations associated with noise, while the peak-to-peak ripple captures low-frequency drift or oscillation. Because constant force signals lack frequency content, conventional frequency-domain measures, reduce to the DC component and therefore fail to reveal the actual steady-state characteristics. For sinusoidal and composite force inputs, the amplitude–frequency response ($A(\omega)$) and the total frequency content (TFC) are employed. The measure $A(\omega)$ quantifies the fidelity of the output amplitude with respect to the input, whereas TFC characterizes the distribution and energy of frequency components. Together, these indices provide sensitivity to the actuator's dynamic adaptability and frequency-domain response capability.

In terms of transient response, settling time, root mean square error (RMSE), and integral of absolute error (IAE) are selected as the principal performance indicators. Settling time reflects the convergence speed, RMSE captures the overall deviation magnitude, and IAE emphasizes the accumulation of error over time. The combination of these three metrics enables a comprehensive evaluation of control accuracy and stability.

4. Experimental results and discussion

This section evaluates the EM-KF-RTS algorithm for force identification in detail and applies it to three different actuators, ASCPA, LFPA and TPA, to verify its performance in a grinding force control scenario. The experiments were conducted in three undisturbed and two disturbed grinding scenarios, so as to systematically evaluate the identification ability of the algorithm and the dynamic performance of the actuators. In the actuator comparison experiment, in order to ensure fairness, a unified control and force estimation framework was adopted: all three actuators used the same force controller (reference [56]) and the same EM-KF-RTS-based force identification module. The only factor that was changed was the hardware of the actuator. Therefore, the observed differences in force tracking behavior and force signal quality can be attributed to the structural characteristics of the actuator, rather than changes in the controller or signal processing flow.

All experiments were conducted using a closed-loop force controller. The force controller used in the experimental platform is derived from our previous research (Ref [58]), which developed and experimentally validated an observer-based adaptive neural force tracking controller for pneumatic end effectors with actuator saturation. In this study, this controller was used for real-time adjustment of the grinding force, and the main objective was to improve the quality of force feedback under noisy measurement conditions using the proposed EM-KF-RTS force identification method. Specifically, $F_d(k)$ represents the desired contact force, and $F_m(k)$ represents the original force sensor measurement. The proposed recognition module processes $F_m(k)$ to generate a denoised estimated force $\hat{F}(k)$. The force tracking error is defined as $e_F(k) = F_d(k) - \hat{F}(k)$. Therefore, compared with the conventional feedback using $F_m(k)$, the control loop in this paper is implemented by replacing the direct measurement feedback with $\hat{F}(k)$, so that the controller operates with a noise-suppressed force feedback signal. For completeness, the valve control input applied in the experiments follows Eq. (75) in Ref. [58] and can be written as

$$u(k) = -(k_3 + 0.5)z_3(k) - \frac{1}{2a_3^2}z_3(k)\hat{\phi}_3(k)W_3^T(z_3(k))W_3(z_3(k)) \quad (12)$$

The definitions of $z_3(\cdot)$, $W_3(\cdot)$, and $\hat{\phi}_3(\cdot)$ can be found in Ref. [58]. Detailed controller synthesis and stability analysis will not be elaborated upon here, as they are not the focus of this paper; interested readers can refer to Ref. [58] for complete derivations and proofs.

From an implementation perspective, the computational cost and real-time feasibility of integrating the EM-KF-RTS framework into grinding force regulation require detailed analysis. The closed-loop controller needs an online force feedback signal, provided by a causal Kalman filtering stage that can be executed in real-time on a sample-by-sample basis. However, the standard Rauch-Tung-Striebel smoother is inherently non-causal because it utilizes future measurements; therefore, when real-time operation is required, smoothing is performed using a fixed-delay or short-delay window approach, which introduces limited latency while improving noise suppression. To further control time costs, the expectation maximization (EM) parameter adaptively employs a finite number of iterations per window and reduces the update rate when necessary, rather than iterating to full convergence at every sample. Furthermore, the harmonic state-space representation used has a compact state dimension, thus the associated matrix operations remain lightweight and computationally inexpensive. With these implementation settings, utilizing causal filtering for feedback, and window-based smoothing and adaptation for enhancement, with bounded delays, the entire identification and control process is compatible with real-time grinding force regulation under noisy measurements.

Case 1: Constant force signal $y_r = 35\text{N}$

Constant-force grinding represents a typical precision machining scenario, where vibration, sensor noise, and electromagnetic interference inevitably corrupt the measured force signals. In this study, a reference grinding force of $y_r = 35\text{N}$ was selected. As shown in Fig. 13(a), the raw sensor force signal exhibits significant noise. By applying the proposed EM-KF-RTS algorithm, the force signal was effectively identified and denoised, enabling accurate input for the adaptive controller to track the reference force. From the force tracking results (Fig. 13(b)) and error trajectories (Fig. 13(c)), it is evident that the EM-KF-RTS algorithm enables more accurate reconstruction of the true grinding force, thereby facilitating precise tracking across all three actuators. Notably, the ASCPA exhibits the best overall performance.

After analyzing the experimental data, ASCPA achieved the shortest settling time (6.124 s), significantly outperforming TPA (19.244 s) and LFPA (10.400 s). In terms of tracking accuracy, ASCPA also yielded the lowest RMSE (0.087 N) and IAE (1.060 N·s), indicating superior dynamic tracking capability. These improvements arise from structural differences: TPA employs rubber seals that introduce substantial sliding friction, while LFPA adopts a clearance seal with ball guides and a metallic sealing layer, which reduces but does not eliminate intermittent contact. In contrast, ASCPA integrates hydrostatic air-film support with non-contact air bearing technology, establishing a stable air film that eliminates solid contact at sealing interfaces and minimizes frictional losses, thereby

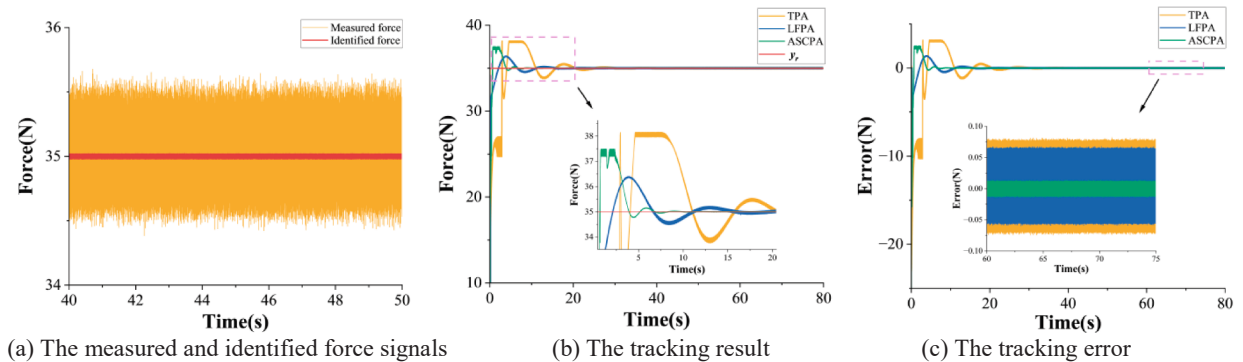


Fig. 13. Experimental results for the constant force signal with $y_r = 35\text{N}$.

enabling faster response. Evaluation of force signal recognition performance, ASCPA further exhibited the smallest steady-state deviation ($\sigma = 0.001$ N) and peak-to-peak ripple ($F_{pp} = 0.003$ N), both markedly lower than those of TPA and LFPA. This improvement results from the synergy between the EM-KF-RTS identification algorithm and ASCPA's air-film structure: the algorithm reconstructs the true force signal from noisy measurements, while the air-film effectively attenuates vibration. Together, these factors enhance signal fidelity and confirm the robustness of the proposed identification framework.

Case 2: Sinusoidal force signal $y_r = \sin(t) + 35$ N

To further evaluate force identification under time-varying grinding loads, a sinusoidal reference signal $y_r = \sin(t) + 35$ N was applied. As shown in Fig. 14(a), the raw sensor measurements contained substantial noise, whereas the EM-KF-RTS algorithm effectively reconstructed the underlying force, yielding a more accurate representation of the true grinding force. The tracking results in Fig. 14(b) show that, with force identification, all three actuators were able to follow the time-varying force within finite time. Among them, ASCPA achieved a response time of 5.318 s, which is 43.6% faster than LFPA (9.417 s) and 59.5% faster than TPA (13.132 s). This improvement mitigates the overshoot and phase lag typically observed under high-frequency force fluctuations. Correspondingly, As shown in Fig. 14(c), ASCPA obtained the lowest tracking error, with an RMSE of 0.089 N, representing a 7.3% reduction compared with LFPA (0.096 N) and a 63.4% improvement relative to TPA (0.243 N). The superior performance of ASCPA arises from its reduced pneumatic hysteresis, which enables more stable and responsive force regulation.

Through frequency-domain analysis of the experimental data, the amplitude–frequency response ($A(\omega)$) and the total frequency content (TFC) were examined. ASCPA exhibited $A(\omega) = 0.348$ and a TFC of 55.712%. The lowest $A(\omega)$ indicates its ability to suppress excessive amplification at the driving frequency while retaining the dominant spectral content, thereby improving tracking stability and reducing susceptibility to high-frequency disturbances. Moreover, the lowest TFC demonstrates that ASCPA yields the most spectrally compact response, reflecting reduced harmonic distortion and noise leakage.

Case 3: Variable frequency amplitude force signal $y_r = \sin(3t) - \cos(2t) + 35$ N

To further evaluate the performance of the proposed EM-KF-RTS algorithm under more complex grinding conditions, a variable frequency amplitude force signal $y_r = \sin(3t) - \cos(2t) + 35$ N was applied. As shown in Fig. 15(a), although measurement noise increases under such conditions, the EM-KF-RTS framework consistently extracts the true grinding force, enabling all three pneumatic actuators to achieve accurate force tracking. This robustness stems from the algorithm's data-driven EM optimization, which adaptively identifies state-space parameters and effectively suppresses noise leakage. The tracking results in Fig. 15(b) demonstrate that ASCPA achieved the fastest response time of 2.732 s, significantly outperforming LFPA (9.878 s) and TPA (16.987 s). This improvement is attributed to the hydrostatic air-film suspension combined with non-contact air bearing design, which eliminates sealing friction and mechanical hysteresis. In contrast, LFPA suffers from residual sliding friction, while TPA is limited by rubber seal deformation and stick–slip effects, leading to higher sensitivity to high-frequency load fluctuations. The error trajectories in Fig. 15(c) confirm that ASCPA attained the lowest RMSE (0.132 N), effectively suppressing overshoot and oscillations induced by force magnitude and frequency transitions.

Frequency-domain analysis of the amplitude–frequency response ($A(\omega)$) and total frequency content (TFC) demonstrates that all three pneumatic actuators achieve favorable spectral characteristics under the proposed EM-KF-RTS framework. The algorithm effectively attenuates high-frequency noise while preserving dominant spectral components, thereby enabling a more faithful reconstruction of the actual grinding force across all cases. Among the three actuators, ASCPA exhibits lowest amplitude ratios ($A(\omega_1) = 0.325$ at $\omega = 3$ rad/s and $A(\omega_1) = 0.357$ at $\omega = 2$ rad/s) and the most stable spectral distribution (TFC = 29.862%). This behavior arises from its hydrostatic air-film support and non-contact air bearing design, which suppress mechanical friction and isolate structural vibrations, thereby reducing the propagation of spurious spectral energy. The resulting lowest in $A(\omega)$ and TFC variation highlight superior noise immunity and spectral robustness, which are particularly advantageous for mitigating force overshoot and

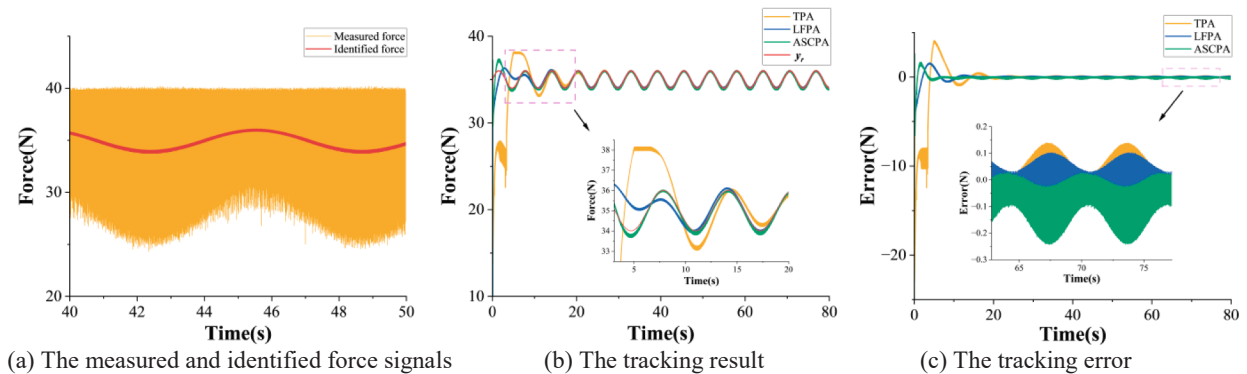


Fig. 14. Experimental results for the sinusoidal force signal with $y_r = \sin(t) + 35$ N.

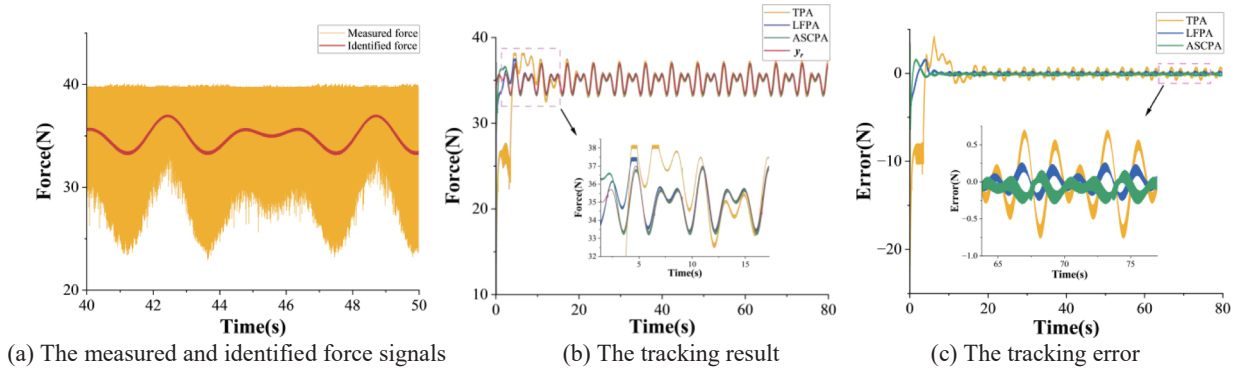


Fig. 15. Experimental results for the variable frequency amplitude force signal with $y_r = \sin(3t) - \cos(2t) + 35N$

phase lag during practical grinding operations.

To quantitatively compare the performance of the three pneumatic actuators and to assess the effectiveness of the proposed EM-KF-RTS algorithm under different grinding force conditions, several performance indices were selected, including settling time, root mean square error (RMSE), integral of absolute error (IAE), steady-state standard deviation (σ), peak-to-peak ripple (F_{pp}), amplitude-frequency response ($A(\omega)$), and total frequency content (TFC). The results are summarized in Table 4. Statistical analysis clearly demonstrates the superior force identification capability of the proposed EM-KF-RTS algorithm across varying force signals, as well as the outstanding tracking performance of the designed ASCPA structure.

Case 4: External interference experiment

To further validate the force identification capability of the proposed EM-KF-RTS algorithm under unknown external disturbances, as well as the disturbance-rejection capacity of the ASCPA structure. After achieving constant force tracking, abrupt external contact perturbations were introduced to generate irregular disturbances, thereby enabling a comprehensive evaluation of actuator performance.

As shown in Fig. 16(a), under abrupt external disturbances the proposed EM-KF-RTS algorithm also effectively identifies the true grinding force, enabling all three pneumatic actuators to regain accurate tracking. The magnified view highlights that ASCPA exhibits the fastest recovery and the smallest force fluctuations, whereas TPA shows the slowest recovery with pronounced oscillations. Fig. 16(b) further illustrates the error trajectories, where ASCPA consistently maintains small-amplitude fluctuations throughout the disturbance period, demonstrating superior stability. The PSD comparison in Fig. 16(c) reveals that ASCPA, benefiting from its non-contact lubrication and low-disturbance characteristics, achieves the lowest and smoothest error energy distribution during recovery. LFPA provides intermediate suppression, while TPA suffers from high total error energy and evident resonant components, indicating the poorest dynamic recovery and stability. Finally, Fig. 16(d) quantitatively summarizes these observations. The ASCPA achieves the lowest integral of absolute error (IAE), the shortest adjustment time, and the smallest peak deviation, indicating that the proposed structure effectively suppresses force oscillations and thereby minimizes the risks of over-grinding and instability.

In addition to the abrupt contact impact, a differential pressure variation was introduced as an external disturbance source during the sinusoidal force tracking experiment. Specifically, after the three pneumatic actuators achieved steady-state tracking, irregular disturbances were generated by altering the pressure differential between the two chambers of the pneumatic actuators.

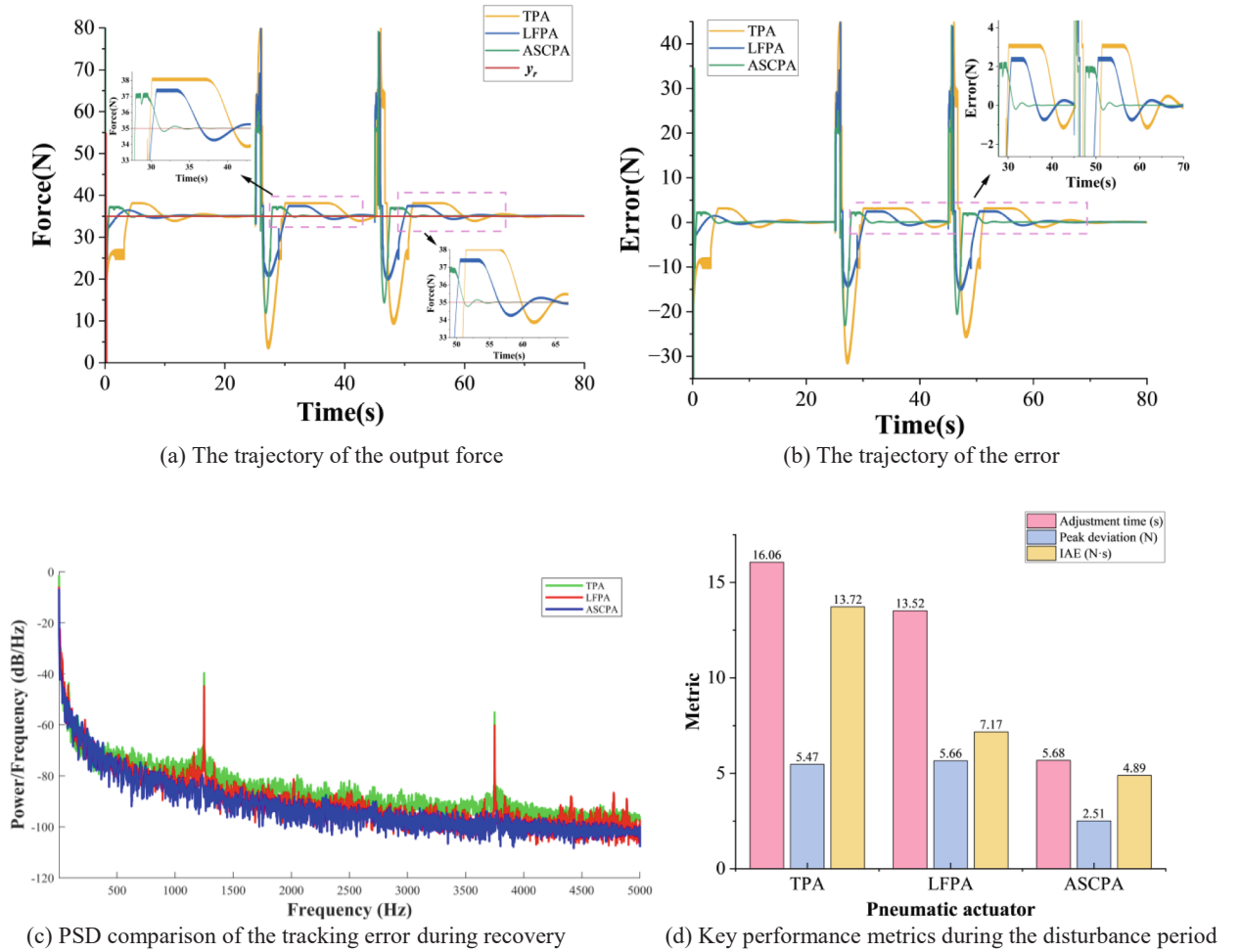
As shown in Fig. 17(a) and (b), the output force trajectories and corresponding error profiles indicate that, owing to the accurate reconstruction of the true grinding force by the proposed EM-KF-RTS algorithm, all three actuators were able to reestablish stable tracking despite pressure fluctuations. Among them, the ASCPA exhibited the shortest adjustment time and the highest control accuracy. By contrast, the TPA suffered from larger oscillations during recovery, primarily due to higher sealing friction and compliance-induced hysteresis of its rubber sealing elements. The PSD comparison of tracking errors in Fig. 17(c) shows that power spectral density of all actuators decreases with increasing frequency. However, the TPA presents the highest overall PSD level with pronounced peaks near 1300 Hz and 3800 Hz. These peaks originate from friction-induced stick-slip dynamics and structural resonance associated with compliant seals. In contrast, the ASCPA achieves the lowest and flattest PSD distribution, attributable to the contactless hydrostatic air-film support, which effectively suppresses frictional excitation and attenuates the transmission of high-frequency disturbances. As shown in Fig. 17(d), among the three actuators, the ASCPA achieved the smallest adjustment time, peak deviation, and integral of absolute error (IAE). These results demonstrate that the combination of the ASCPA structure with the EM-KF-RTS force recognition algorithm enables rapid adjustment and effective suppression of force oscillations under external disturbances, thereby minimizing risks of improper operation and significantly enhancing process stability in practical applications.

In summary, the proposed ASCPA structure consistently outperforms TPA and LFPA across all three disturbance-free force-tracking scenarios and two disturbance-imposed grinding conditions, exhibiting superior dynamic response, tracking accuracy, and robustness. Meanwhile, the EM-KF-RTS algorithm enables reliable noise suppression and precise force identification in both steady and perturbed environments. These findings collectively confirm the strong potential of the proposed framework for practical application in complex

Table 4

Comparison of performance indicators.

Case	$y_r(t)$	Pneumatic actuator	Settling time(s)	RMSE (N)	IAE (N·s)	$\sigma(N)$	$F_{pp}(N)$	$A(\omega)$	TFC
Case1	35N	TPA	19.244	0.237	5.732	0.006	0.046	/	25.851%
		LFPA	10.400	0.097	1.271	0.003	0.015	/	25.855%
		ASCPA	6.124	0.087	1.060	0.001	0.003	/	25.851%
Case2	$\sin(t) + 35N$	TPA	13.132	0.243	5.516	0.240	6.161	3.178	55.713%
		LFPA	9.417	0.096	1.338	0.113	5.993	0.662	55.721%
		ASCPA	5.318	0.089	1.070	0.116	6.194	0.348	55.712%
Case3	$\sin(3t) - \cos(2t) + 35N$	TPA	16.987	0.263	11.630	0.260	6.184	2.817	29.866%
		LFPA	9.878	0.136	7.064	0.134	6.179	0.516	29.862%
		ASCPA	2.732	0.132	6.769	0.133	6.179	0.325	29.862%

**Fig. 16.** Experimental results of constant force tracking subject to abrupt external contact disturbances.

grinding processes.

To further verify the practical effectiveness of the proposed method in material removal scenarios, we conducted actual grinding experiments on a typical curved workpiece using the end effector of the developed grinding system. Two adjacent regions on the same curved surface specimen were machined under identical grinding conditions, including the same motion path, feed conditions, and force controller. The only difference between the two regions was the feedback configuration of the force loop: Region A used the force identified based on the EM-KF-RTS method as the feedback signal, while Region B used the original force feedback. This design minimizes the differences between specimens and allows for direct comparison of the impact of force identification on the final surface finish. Surface roughness was evaluated using the arithmetic mean roughness R_a . As shown in Fig. 18, eight measurement points were

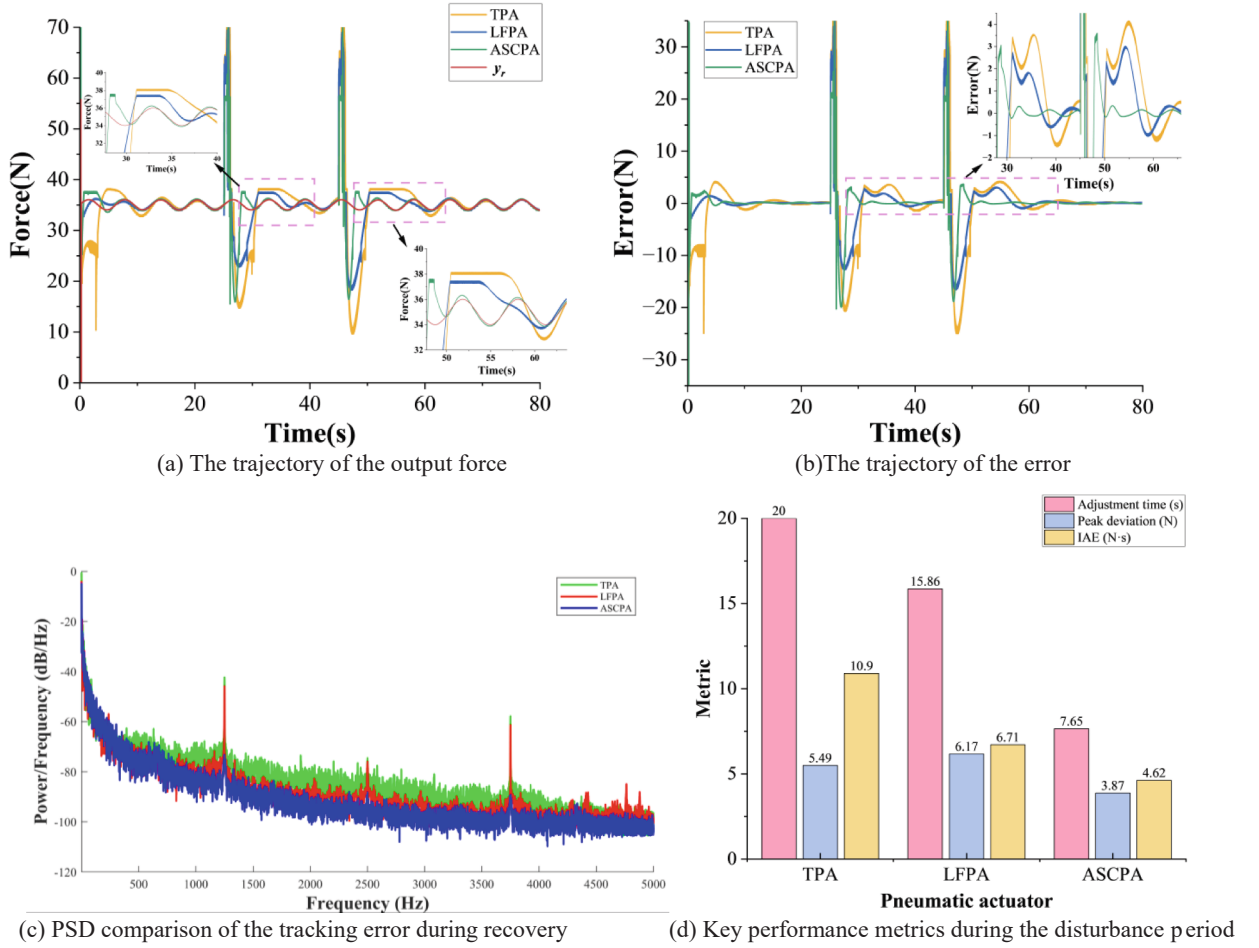


Fig. 17. Experimental results of constant force tracking subject to disturbances from pressure differential variation.

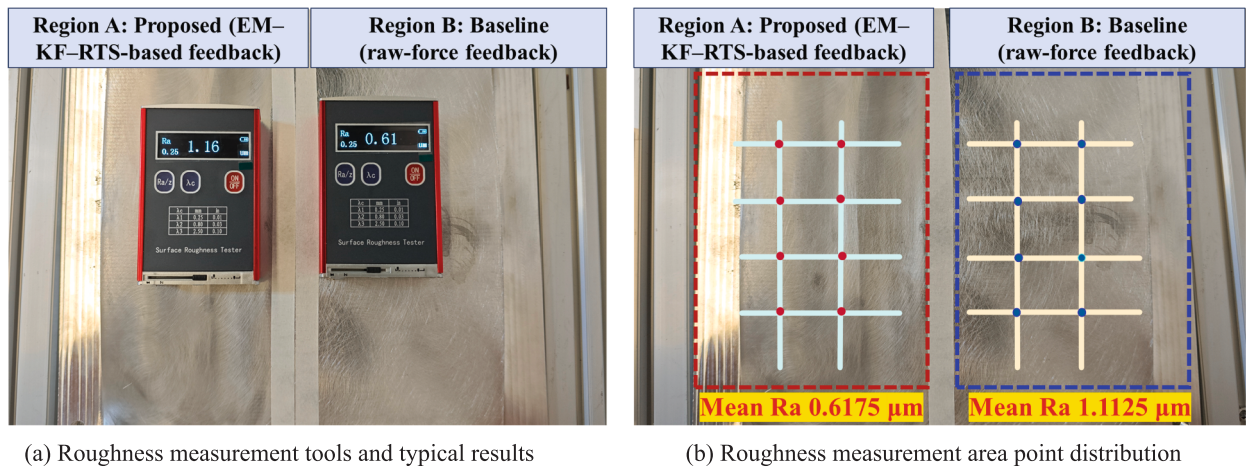


Fig. 18. Actual grinding validation on curved workpiece and roughness measurement layout under two feedback configurations.

distributed in each machining region to cover different local contact locations within the grinding area. Roughness readings were collected at each measurement point, and the results are summarized in Table 5. The Ra values of the baseline region were 1.15, 1.00, 1.16, 1.03, 1.12, 1.09, 1.21, and 1.14 μm , with an average of 1.1125 μm and a coefficient of variation (CV) of 6.35%. In contrast, the Ra

Table 5

Surface roughness measurement results for actual grinding of curved workpiece.

Region A Ra	Mean Ra	CV%	Region B Ra	Average Ra	CV%
0.63 μm	0.6175 μm	4.72%	1.15 μm	1.1125 μm	6.35%
0.58 μm			1.00 μm		
0.64 μm			1.16 μm		
0.59 μm			1.03 μm		
0.61 μm			1.12 μm		
0.60 μm			1.09 μm		
0.67 μm			1.21 μm		
0.62 μm			1.14 μm		

values of the proposed region were 0.63, 0.58, 0.64, 0.59, 0.61, 0.60, 0.67, and 0.62 μm , with an average of 0.6175 μm and a CV of 4.72%. Compared to the baseline configuration, the proposed EM-KF-RTS-based feedback method reduced the average surface roughness by 44.5% while also reducing the spatial dispersion of roughness between measurement points, decreasing the CV from 6.35% to 4.72%. The lower average Ra value indicates improved surface finish, while the reduced CV value indicates more consistent material removal behavior within the machined curved surface region. These results provide manufacturing-related evidence that replacing the original force feedback with the force identified by EM-KF-RTS in a closed-loop controller can improve grinding performance under noise measurement conditions.

5. Conclusion

This study proposes the design of a novel aerostatic suspension compact pneumatic actuator (ASCPA), developed to achieve low-friction performance for compact pneumatic end-effectors in grinding applications. Based on static-pressure air-suspension lubrication, fluid dynamic simulations were conducted to optimize piston parameters affecting radial load capacity and air consumption, guiding prototype fabrication. A dedicated tribological test platform and comparative experiments against a TPA and an LFPA confirmed the ASCPA's superior low-friction characteristics. To ensure accurate extraction of dynamic grinding forces under strong noise arising from vibration, electromagnetic interference, and sensor limitations, an adaptive state-space optimal estimation algorithm (EM-KF-RTS) is further proposed. This framework integrates Kalman filtering and Rauch-Tung-Striebel smoothing, with parameter adaptation via the Expectation-Maximization method, enabling robust noise suppression and precise force recognition beyond conventional frequency-domain approaches. A dynamic grinding testbed was established to evaluate performance under three disturbance-free conditions (constant force, sinusoidal force, and variable frequency-amplitude force) and two disturbance scenarios (abrupt external contact and inter-chamber pressure variation). Results demonstrate that the EM-KF-RTS algorithm achieves reliable noise suppression and accurate force identification across all conditions, while the ASCPA consistently outperforms TPA and LFPA in response speed and tracking accuracy. Together, the proposed actuator-algorithm framework offers a promising solution for high-precision, disturbance-resilient force control in pneumatic grinding applications.

CRedit authorship contribution statement

Zhiguo Yang: Writing – original draft. **Zhanxin Li:** Formal analysis. **Yanxia Niu:** Software. **Jiange Kou:** Validation. **Xiangkai Shen:** Data curation. **Yixuan Wang:** Funding acquisition. **Zhibo Sun:** Project administration. **Yushan Ma:** Project administration. **Yan Shi:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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